

The ODE Template and Its Domain Equivalents

Mechanical and electrical forms of one second-order equation

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Abstract

Three linear relations and one balance produce the template $a\ddot{y} + b\dot{y} + cy = 0$. Substituting coefficients and coordinates gives four forms: MCK, KCM, LRC and CRL. We derive their mappings, the reciprocal substitution between dual forms, and the scales that make the mechanical and electrical equations correspond exactly. The forced template also fixes the harmonic convention used by the companions.

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Latest on GitHub: <https://github.com/hobnilre/physics-ode-template>

1 The common template

Let $a, b, c > 0$ be constants and $D = d/dt$; dots and primes also denote time derivatives. For a common quantity u , define coordinates $\dot{y} = u$, $h = au$, and three terms of the same dimension:

$$z_a = \dot{h} = a\dot{u}, \quad z_b = bu, \quad z_c = cy. \quad (1)$$

Choose their signs so that the unforced balance is $z_a + z_b + z_c = 0$. Substitution gives

$$\boxed{a\ddot{y} + b\dot{y} + cy = 0.} \quad (2)$$

The coefficients multiply the rate of u , u itself, and its accumulated coordinate. Their meanings determine the domain.

Write T for the time dimension and $[z]$ for the common dimension of the three terms. The defining relations require

$$[y] = [u]T, \quad [h] = [z]T, \quad [a] = \frac{[z]T}{[u]}, \quad [b] = \frac{[z]}{[u]}, \quad [c] = \frac{[z]}{[u]T}. \quad (3)$$

The equivalent first-order system is

$$Dy = \frac{h}{a}, \quad Dh = -cy - \frac{b}{a}h, \quad (4)$$

with initial coordinates $y(t_0)$ and $h(t_0) = a\dot{y}(t_0)$.

1.1 Forcing and harmonic form

A prescribed sum gives $P_2(D)y = f$, $P_2(s) = as^2 + bs + c$. Phasors use $x(t) = \text{Re}(\hat{x}e^{i\omega t})$, $i^2 = -1$ and $\omega > 0$. Thus D multiplies a phasor by $i\omega$, and

$$\hat{f} = P_2(i\omega)\hat{y} = \left(ai\omega + b + \frac{c}{i\omega}\right)\hat{u}, \quad \hat{u} = i\omega\hat{y}. \quad (5)$$

The full solution retains its initial coordinates. Higher-order coefficients and their phasor factors are derived in *ODE Coefficient Synthesis*.

2 Four domain forms

In mechanics, let m be mass, d the damper coefficient and k stiffness; use displacement x , velocity v , force F and momentum p . In electrics, use inductance L , resistance R , capacitance C , current i , voltage e , charge q and flux linkage λ . All parameters are positive constants.

Table 1. The coefficient and quantity dictionary. Here x_s is spring displacement; the last row identifies the quantity being summed.

Template	MCK	KCM	LRC	CRL
a	m	$1/k$	L	C
b	d	$1/d$	R	$1/R$
c	k	$1/m$	$1/C$	$1/L$
Common u	v	F	i	e
Coordinate $h = au$	p	x_s	λ	q
Coordinate y	x	p	q	λ
Type of z_j	Force	Velocity	Voltage	Current

2.1 MCK and LRC

For MCK, velocity is common and signed forces add. With $v = \dot{x}$ and $p = mv$, the terms are $\dot{p} = m\ddot{x}$, $d\dot{x}$ and kx . For LRC, current is common and signed voltages add. With $i = \dot{q}$ and $\lambda = Li$, the terms are $\dot{\lambda} = L\ddot{q}$, $R\dot{q}$ and q/C . Their balances give

$$m\ddot{x} + d\dot{x} + kx = 0, \quad L\ddot{q} + R\dot{q} + \frac{q}{C} = 0. \quad (6)$$

These are the mechanical parallel and electrical series forms.

2.2 KCM and CRL

For KCM, force is common and signed velocities add. The relations $\dot{p} = F$, $x_s = F/k$, $v_m = p/m$ give velocities $\dot{F}/k$, F/d and p/m . For CRL, voltage is common and signed currents add. The relations $\dot{\lambda} = e$, $q = Ce$, $i_L = \lambda/L$ give currents $C\dot{e}$, e/R and λ/L . Hence

$$\frac{\ddot{p}}{k} + \frac{\dot{p}}{d} + \frac{p}{m} = 0, \quad C\ddot{\lambda} + \frac{\dot{\lambda}}{R} + \frac{\lambda}{L} = 0. \quad (7)$$

These are the mechanical series and electrical parallel forms.

3 Correspondence and duality

3.1 Changing domains

Choose positive conversion scales α, β and use the same time variable. The force–voltage correspondence is

$$i \leftrightarrow \alpha v, \quad e \leftrightarrow \beta F, \quad L = \frac{\beta}{\alpha} m, \quad R = \frac{\beta}{\alpha} d, \quad C = \frac{\alpha}{\beta k}. \quad (8)$$

The scales convert units as well as numerical values. For MCK–LRC, $(q, \lambda) = (\alpha x, \beta p)$; for KCM–CRL, $(\lambda, q) = (\beta p, \alpha x_s)$. Direct substitution gives

$$L\ddot{q} + R\dot{q} + \frac{q}{C} = \beta(m\ddot{x} + d\dot{x} + kx), \quad (9)$$

$$C\ddot{\lambda} + \frac{\dot{\lambda}}{R} + \frac{\lambda}{L} = \alpha \left(\frac{\ddot{p}}{k} + \frac{\dot{p}}{d} + \frac{p}{m} \right). \quad (10)$$

Thus MCK corresponds to LRC, and KCM to CRL; initial coordinates follow the same maps.

3.2 Exchanging common and summed quantities

Construct the dual template by making a quantity w common in place of the z_j , and summing the corresponding quantities u_j . Reversing the relations (1), in reverse element order, gives

$$u_c = \frac{\dot{w}}{c}, \quad u_b = \frac{w}{b}, \quad u_a = \frac{Y}{a}, \quad \dot{Y} = w.$$

Here $Y = au_a$ is the coordinate associated with the original a relation. The balance $u_c + u_b + u_a = 0$ becomes

$$\boxed{\frac{\ddot{Y}}{c} + \frac{\dot{Y}}{b} + \frac{Y}{a} = 0, \quad \mathcal{D}(a, b, c) = \left(\frac{1}{c}, \frac{1}{b}, \frac{1}{a} \right).} \quad (11)$$

Applying $\mathcal{D}$ twice returns (a, b, c) . It sends MCK to KCM and LRC to CRL, exactly as in Table 1.

Dividing each equation by its leading coefficient makes the effect of this substitution explicit:

$$\ddot{y} + \frac{b}{a}\dot{y} + \frac{c}{a}y = 0, \quad \ddot{Y} + \frac{c}{b}\dot{Y} + \frac{c}{a}Y = 0. \quad (12)$$

The ratio c/a is unchanged. The first-derivative coefficient changes from b/a to c/b ; the two monic equations coincide exactly when $b^2 = ac$.

The scales change domains; the reciprocal map constructs the dual. Coupled blocks and their initial coordinates are treated in *Interconnecting Series and Parallel ODEs*; signed power and work are treated in *Energy Ledgers for Forced Harmonic ODEs*.

References

1. H. Nilre and B. C. Herlin (2026). [ODE Coefficient Synthesis](#).
2. H. Nilre and B. C. Herlin (2026). [Interconnecting Series and Parallel ODEs](#).
3. H. Nilre and B. C. Herlin (2026). [Energy Ledgers for Forced Harmonic ODEs](#).